Experiment 12

Memory Allocation Strategies

code:

```
#include <stdio.h>

int main(void) {
   int nb, np, b[20], p[20], choice;
   int i, j, best;

   printf("Number of memory blocks (max 20): ");
   scanf("%d", &nb);

   printf("Number of processes (max 20): ");
   scanf("%d", &np);

   if (nb < 1 || nb > 20 || np < 1 || np > 20)
      return 1;

   for (i = 0; i < nb; i++) {
      printf("Block %d size: ", i + 1);
      scanf("%d", &b[i]);
   }

   for (i = 0; i < np; i++) {
      printf("Process %d size: ", i + 1);
      scanf("%d", &p[i]);
   }

   printf("1 First Fit  2 Best Fit  3 Worst Fit: ");
   scanf("%d", &choice);
```

```
if (choice < 1 || choice > 3)
   return 1;

for (i = 0; i < np; i++) {
   best = -1;

   for (j = 0; j < nb; j++) {
      if (b[j] >= p[i]) {

         if (best == -1 ||
            (choice == 2 && b[j] < b[best]) ||
            (choice == 3 && b[j] > b[best])) {
            best = j;
         }

         if (choice == 1) {
            best = j;
            break;
         }
      }
   }

   if (best == -1) {
      printf("P%d (%d): Not allocated\n", i + 1, p[i]);
   } else {
      printf("P%d (%d) allocated to Block %d\n",
           i + 1, p[i], best + 1);

      b[best] -= p[i];
   }
}
```

```
    return 0;
}
```

Output:

